

Stereochemistry is EZ!

Thanks to Jonah H. for the title.

Organic Chemistry CHE 310, Exam #3

100 painless points!
November 17, 2017

Rules of the road:

1. There are 9 questions on 5 pages, plus a couple of extra credit questions for your amusement.
2. No Notes.
3. No electronic devices of any kind. This includes, but is not limited to the following: PDAs, cell phones, pagers, calculators, mp3 players and multifunctional watches. Students found with these devices at their desk or on their person will fail the exam and be reported to the Dean of Students. w
4. This exam is for the sole use of the student whose name appears below. This exam may not be reproduced in any form without written consent of the instructor.
5. Read the agreement and ***sign below before you begin.***
6. Good Luck!

K.C. Russell

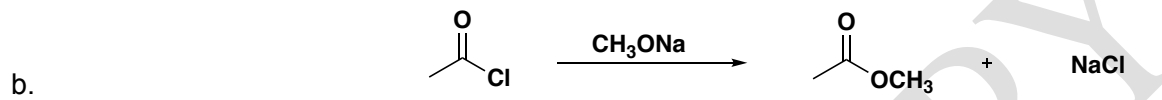
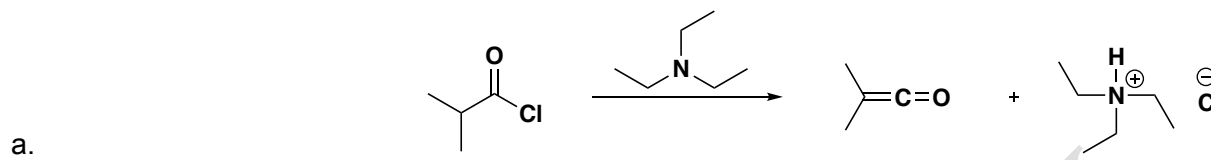
I have read and agree to follow the above rules. Furthermore, I hereby recognize that I am subject to and agree to abide by the *Northern Kentucky University Honor Code*, which provides standards that encourage ethical academic behavior and imposes penalties for violations of such standards.

Printed Name : _____

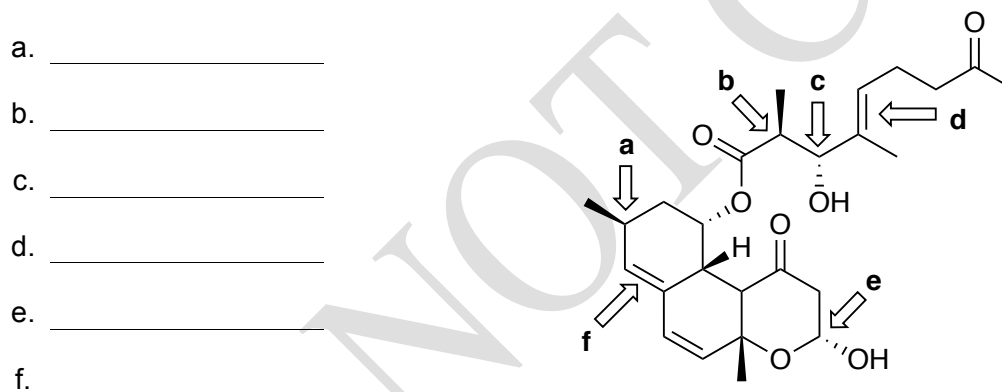
Student ID# _____
Last eight of ISO or last four of SSN

Signature : _____

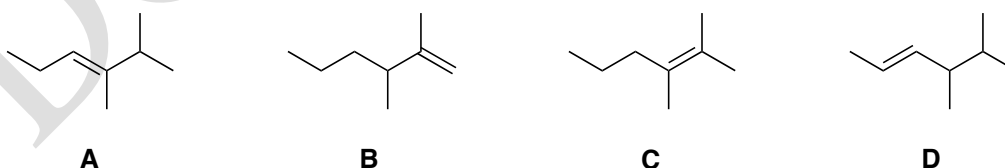
1. Fill in the blank. Classify each reaction as either an addition, elimination, rearrangement, or substitution. (4 points)



2. The compound below was recently isolated from a plant fungus and showed potent activity against colon cancer cell lines. **Assign** the absolute configuration (*R*, *S*, *E*, *Z*) to the indicated sites by **filling** in the blanks. (12 pts)



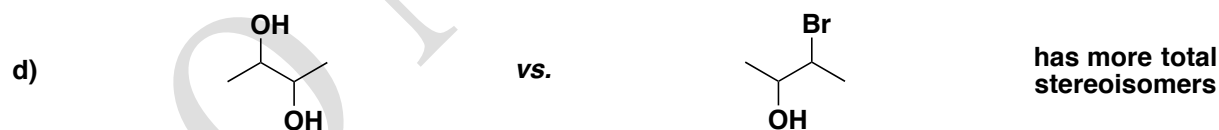
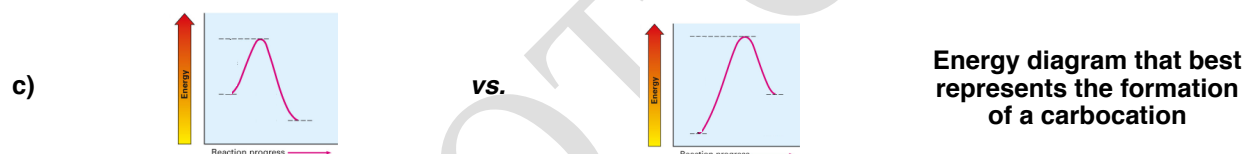
3. Rank the compounds below in the order of decreasing stability (8 points)



_____ > _____ > _____ > _____
 Most stable Least stable

4. For each pair below: (20 points)

- a. **Circle** the molecule that best fits the description on the right. If both molecules fit the description equally circle them both. If neither fits, write “neither.” (2 points)
 b. **Provide** a *brief* explanation regarding the fundamental physical property behind the trait. Typically one sentence will do. (2 points)



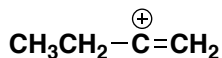
5. Provide the major product(s), missing reagent(s) or starting material(s) for the reactions below. Clearly indicate the stereochemistry in your structures where appropriate. Where more than one stereoisomer is formed you only need to draw one of the stereoisomeric products. Other stereoisomers should be indicated by writing, “+ **enantiomer**” or “+ **diastereomer**”, as appropriate. Check the boxes on the right to indicate whether the reaction product solution would be optically active ($[\alpha]_D \neq 0$) or not optically active ($[\alpha]_D = 0$). (12 points; product(s) and starting material(s) = 2, stereochem = 1)



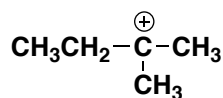
6. Rank the carbocations below in order of increasing stability. (8 pts)



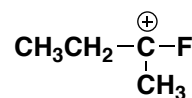
A



B



C



D

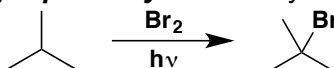
Least stable _____ < _____ < _____ < _____ Most stable

7. Provide the complete mechanism for the reactions below to give the product(s) shown. (20 pts)

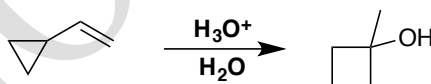
For full credit each mechanism must include the following:

- Proper arrows to show all electron motion.
- The structure of all reactive intermediates including stereochemistry where appropriate.

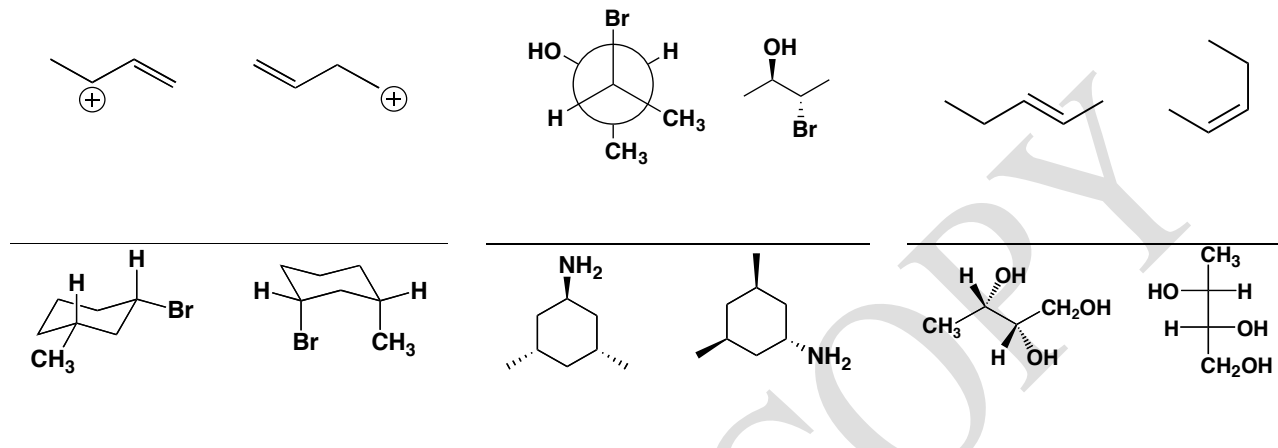
a. This mechanism only needs the **major pathway** necessary to produce the product shown.



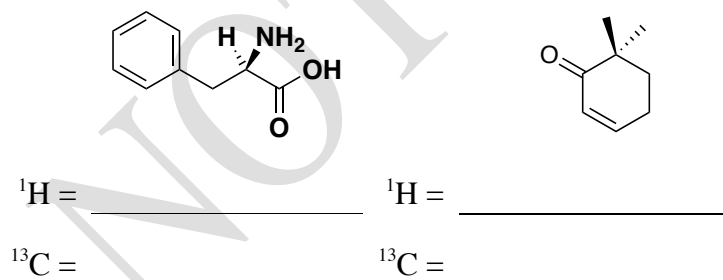
b.



8. For each pair of molecules below fill in the blank with their relationship.
 Choices are **identical**, **enantiomers**, **diastereomers**, **conformational isomers**, **resonance contributors**, **constitutional isomers**, and **non-isomeric**. (12 points)



9. Enter in the blanks the number of peaks that would be observed in the ^1H and ^{13}C NMR spectra of the compounds below. You must consider diastereotopic / enantiotopic groups when solving this problem. (4 points)



Extra Credit.

- In what year did your Laureate win his/her prize? (2 points)
- Which Nobel Laureate received the award for experiments that allowed the examination of transition states? Circle your answer (2 points)

Richard Ernst

George Olah

Vadimir Prelog

Amhed Zewail